

Rapid Microstructure Prediction in Laser Powder Bed Fusion Using Bayesian Updating of Eagar–Tsai Model

Concise subtitle for a Texas A&M proposal, review, or technical talk

Kyle Swartz, Brent Vela, Raymundo Arroyave, Doguhan Sariturk

Texas A&M University

July 7, 2026

Core Thesis

Composition



- A 5-element affine projection represents the accessible composition simplex.
- Interior coloring encodes local element content across the design space.
- Bayesian updates narrow the search to feasible LPBF compositions.

Solidification Condition

- Process parameters determine thermal gradients, cooling rates, and melt-pool shape.
- The Eagar–Tsai model provides fast temperature-field priors across the parameter space.
- These fields define the local solidification path for each candidate build condition.

Solidified Microstructure

- Microstructure emerges from the coupling of chemistry and thermal history.
- Predicted G/R conditions map to expected grain morphology and phase selection.
- The workflow targets rapid microstructure prediction from composition to final structure.

Motivation and Constraints

- Describe the external pressure, mission need, or scientific gap.
- Name the constraints that make the problem nontrivial.
- Explain why existing approaches are too slow, brittle, or expensive.

Technical gap

Existing workflows still need stronger decision support across:

- sparse data,
- competing objectives,
- physics priors,
- experiment risk.

Reusable Two-Column Layout

Stage or category A

- Use this side for requirements, assumptions, or process details.
- Keep bullets short enough to read during a live talk.

Stage or category B

- Mirror the structure on the left when comparison matters.
- Use `\deliverable{}` and `\risk{}` for emphasis.

Downselection rule

Reject options that fail the practical constraints, even if their isolated performance looks strong.

Gradient or Tradeoff Visual

Example design gradient



High-temperature region

Structural region

Replace this annotation with the key takeaway for the figure.

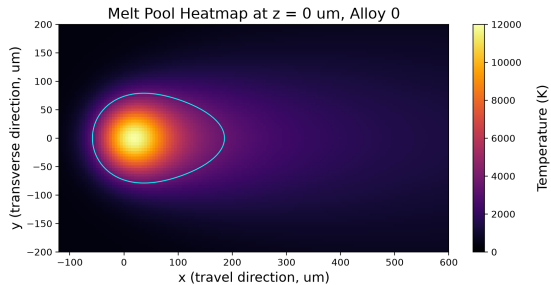
Refactored ET Model

Core idea

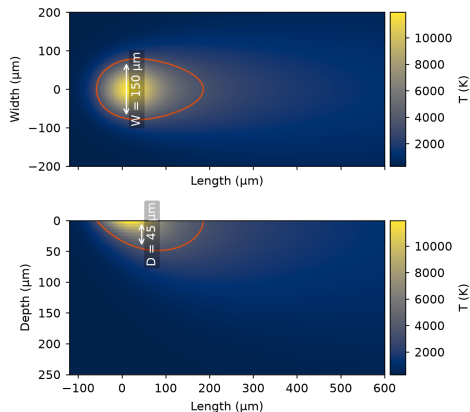
The refactored Eagar–Tsai workflow separates melt-pool prediction, 2D heatmap generation, and 3D temperature-field export into reusable steps.

- Library-backed execution supports direct melt-pool evaluation from process and material inputs.
- Temperature fields can be exported as both dense CSV volumes and VTI files for downstream analysis.
- The workflow is structured for Bayesian correction, microstructure inference, and repeatable parameter studies.

ET Thermal Cross Sections



Built-in refactored ET solver heatmap at $z = 0$ for the 250 W, 0.5 m/s case.



'eagar-tsai' Python library cross section for the same alloy and process condition.

Timeline



Closing

Takeaway

End with the decision, ask, or technical conclusion you want the audience to remember.

- **Deliverable:** Name the concrete output.
- **Primary risk:** Name the hardest unresolved issue.
- **Immediate next step:** Name the action that starts now.